

Foro: Aplicaciones medioambientales de la computación de alto rendimiento (HPC) y la espectroscopia óptica en Centroamérica

Espectroscopia óptica y computacional en estudios de la calidad del agua

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- ① Introduction
- ② Measurements and Simulations
- ③ Results and discussion
- ④ Conclusions and acknowledgements

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Water pollution problems



- † 0.8 M early deaths associated to this type of pollution
- † 1.4% to 2.5% of GDP investment due to water pollution
- †,†† Aguas superficiales contaminadas en El Salvador

Figure: ONU sustainable development goals

† Cost of Environmental Degradation: Air and Water Pollution, World Bank Group, 2021

† Autoridad Salvadoreña del Agua (ASA), laboratorio de calidad del agua, informe 2020

†† Organización Panamericana de la Salud (OPS) y Organización Mundial de la Salud (OMS), Evaluación del sector agua potable y saneamiento en El Salvador; informe preliminar

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Measurement place

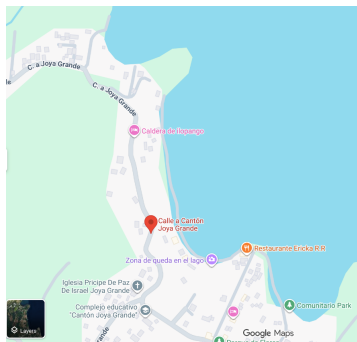
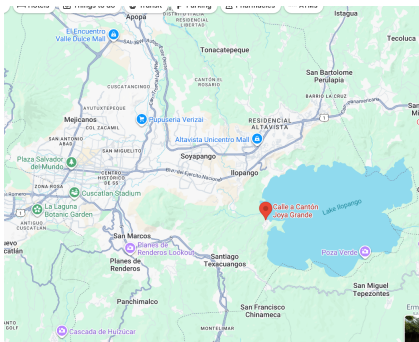


Figure: Cantón Joya Grande, Ilopango lake in map†.

†Screenshot taken with Google Maps.

Measurements - conventional setup

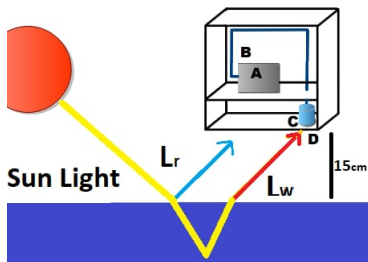


Figure: Diagram for conventional reflectance measurement.



Figure: Conventional Reflectance setup for water quality measurement.

Measurements - MA&Z-DOAS prototype

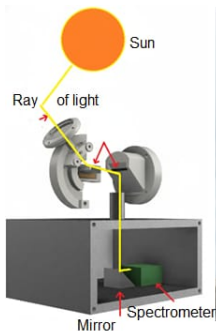
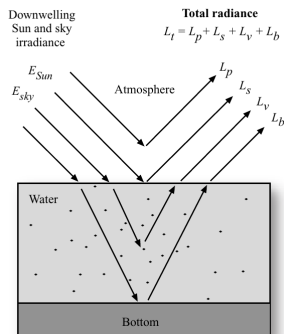


Figure: Diagram for MA&Z-DOAs measurement.



Figure: Conventional Reflectance setup for water quality measurement.

Reflectance model for remote sensing

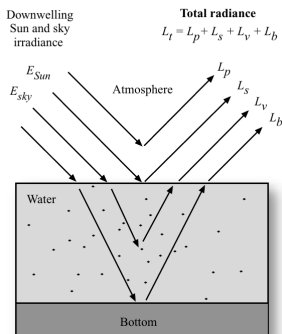


$$R_S(\lambda) = \rho_{ls} \frac{L_t(\lambda)}{E_t(\lambda)} \quad (1)$$

Figure: Diagram of reflected irradiance in a body of water†

†Jensen, J.R. (2007) Remote Sensing of the Environment: An Earth Resource Perspective. 2nd Edition, Pearson Prentice Hall, Upper Saddle River.

Low altitude and Shallow waters approximations



Approximation low altitude:

$$E_{sky} \approx 0$$

$$L_p \approx 0$$

Approximation due to shallow waters:

$$L_s \approx L_v$$

Figure: Diagram of reflected irradiance in a body of water†

†Jensen, J.R. (2007) Remote Sensing of the Environment: An Earth Resource Perspective. 2nd Edition, Pearson Prentice Hall, Upper Saddle River.

Irradiance upwards:

$$L_t = L_{sv} + L_b$$

$$L_b = f_{rs}(\lambda) \times u(\lambda) \times \left\{ 1 - A_{rs,1} \times e^{-[K_d(\lambda) + K_{ub}(\lambda)]z} \right\} \quad (2)$$

$$L_{sv} = A_{rs,2} \times u(\lambda) \times e^{-[K_d(\lambda) + K_{us}(\lambda)]z} \quad (3)$$

Attenuation and retrodispersion:

$$u(\lambda) = \frac{b_b(\lambda)}{a(\lambda) + b_b(\lambda)} \quad (4)$$

$$a(\lambda) = a_w(\lambda) + a_{ph}(\lambda) + a_{CDOM}(\lambda) + a_d(\lambda) \quad (5)$$

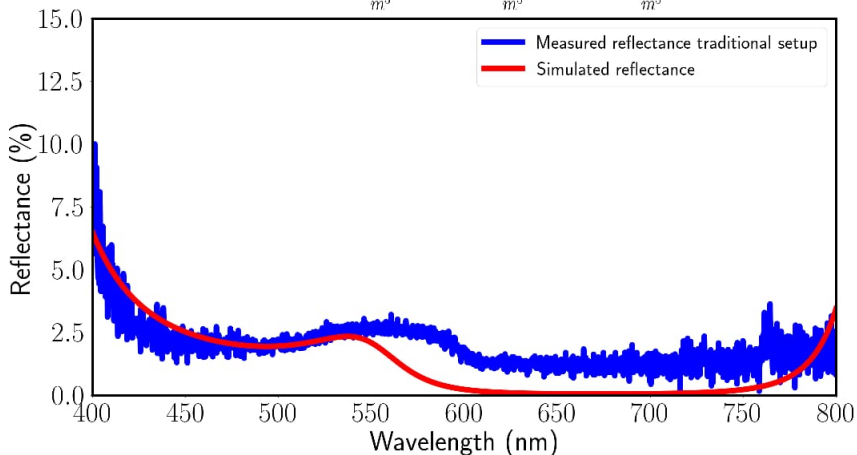
$$b_b(\lambda) = b_{b,w}(\lambda) + b_{b,SM}(\lambda) \quad (6)$$

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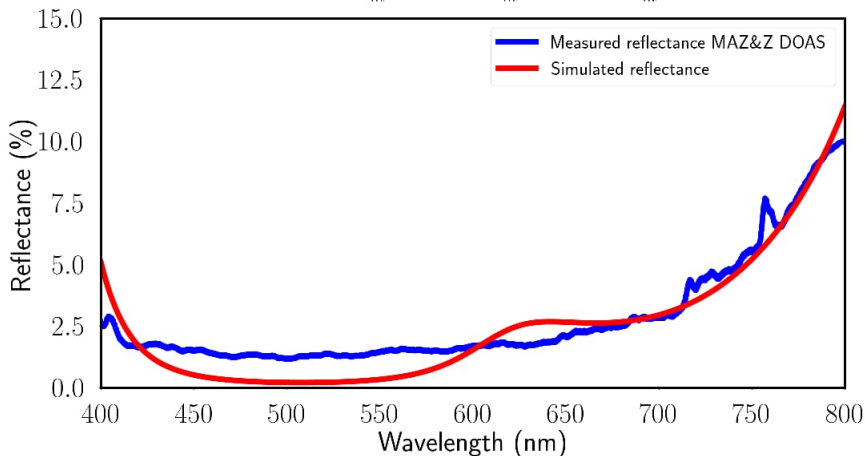
Simulations for traditional setups

$$\begin{aligned} f_{rs} &= 122.90 & c_{ph1} &= 0.41 \frac{\mu g}{m^3} & c_{ph2} &= 66.03 \frac{ng}{m^3} & c_d &= 2.15 \frac{\mu g}{m^3} \\ c_{cdom} &= 16.30 \frac{\mu g}{m^3} & c_{sm} &= 0.55 \frac{ng}{m^3} & c_{mie} &= 9.21 \frac{\mu g}{m^3} \end{aligned}$$



Simulations for MAZ&-DOAS

$$\begin{aligned}f_{rs} &= 125.15 \quad c_{ph1} = 4.88 \frac{\mu g}{m^3} \quad c_{ph2} = 0.41 \frac{ng}{m^3} \quad c_d = 25.77 \frac{\mu g}{m^3} \\c_{cdom} &= 142.72 \frac{\mu g}{m^3} \quad c_{sm} = 0.28 \frac{ng}{m^3} \quad c_{mie} = 9.69 \frac{\mu g}{m^3}\end{aligned}$$



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Conclusions

- Inversion of values for phytoplankton and CDOM in water
- Simulation reproduced quite well the experimental measurements
- MA&Z-DOAS has potential as a low-cost platform for water quality measurements
- Possibility for collaboration for massive analysis data in HPC calculations LARCAD

Acknowledgements

- Consejo Superior Universitario Centroamericano and the International Development Research Centre (CSUCA-IDRC) for partially funding this research under grants 50ES and 45CR
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